COURSE 2 · VIDEO B2 · WORKED SOLUTION

Total Target Cost measure

Video B2 · First steps: Power Query, model, DAX

DESIGN QUESTION

What does one unit of each product cost in parts, at target cost, and which part drives that cost?

Steps

1. File > Save as > Engxtech_B2_Exercise. Home > Transform data > Queries pane > Parts. The TargetCost_USD type icon must show 1.2 (Decimal number). Close & Apply.
2. Model view: you should see two lines into BOM, both one-to-many (1 on Parts and Products, * on BOM), single direction.
3. Report view > Data pane > right-click BOM > New measure. Type the measure below and press Enter.
4. Measure tools > Format: Currency, $ English (United States), 2 decimal places.
5. Card: Total Target Cost. Table 1: Products[ProductName] + Total Target Cost. Table 2: Parts[PartName] + Total Target Cost, sort by the measure (descending).

DAX measure (BOM table)

```
Total Target Cost =
SUMX (
    BOM,
    BOM[QtyPer] * RELATED ( Parts[TargetCost_USD] )
)
```

SUMX walks the BOM rows one at a time. For each row, RELATED follows the many-to-one relationship from BOM to Parts and returns that part's target cost; the row result is QtyPer x cost, and SUMX adds them up. A slicer or a row in a table filters BOM first, so the same measure gives the per-product and per-part numbers.

Why the Drive Motor leads: it costs 85.00 USD at target and sits in two products (CP-100 and MV-300), so its line is 2 x 85.00 = 170.00 USD. The booster pump set has no motor in its BOM, which is why it is the cheapest.

Expected result

Check	Result
Total Target Mass (control: should still read)	**14,865**
Total Target Cost, all products (USD)	**412.00**
Centrifugal pump (CP-100)	**129.50**
Booster pump set (BP-200)	**97.00**
Motorised control valve (MV-300)	**185.50**
Largest part line: Drive Motor (USD)	**170.00**

Common mistakes: SUM (Parts[TargetCost_USD]) ignores QtyPer (two seat rings in a pump count once); a measure written on Parts instead of BOM still works with SUMX (BOM, …), but keep measures next to the table they iterate.